Advanced Biotechnology

Lorenzo Di Michele
Professor of Bionanoscience
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UNIVERSITY OF CAMBRIDGE

Lecture 9

Bioprocessing overview

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Cell removal (part I)



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Lecture outline

1) Overview of bioprocessing
2) (More detailed) overview of downstream processing
3) Principles of filtration for cell removal
4) Mathematical description of dead-end filtration

Chapter 11 Bioprocess Engineering Principles (11.1-11.3)

Learning outcomes

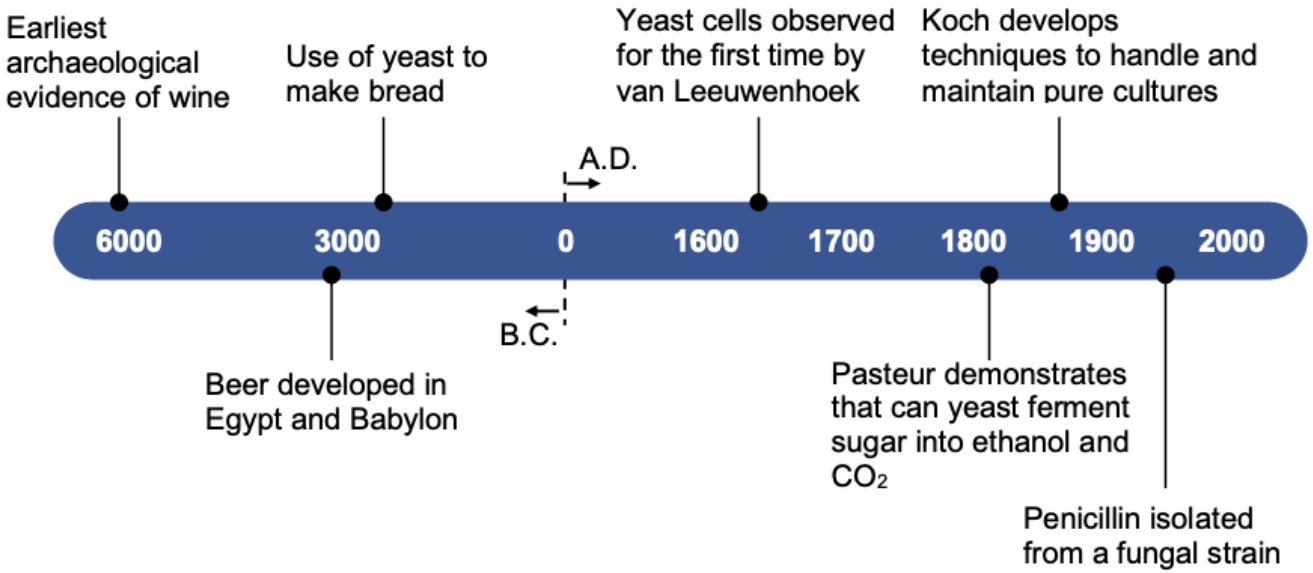
1) Definition of bioprocessing
2) Describe key steps in a bioprocessing pipeline (organism design, scaling up, upstream processing and downstream processing)
3) Define downstream processing and discuss all its key stages
4) Discuss principles and limitations of dead-end filtration for cell removal
5) Mathematically describe dead-end filtration

Lecture outline

1) **Overview of bioprocessing**
2) (More detailed) overview of downstream processing
3) Principles of filtration for cell removal
4) Mathematical description of dead-end filtration

What is bioprocessing?

- Bioprocessing is the use of biological catalysts in the form of live cells (bacteria, plants, fungi, animal cells) or their constituents (enzymes), for the production or transformation of a product
- It is an essential component to many food, chemical and pharma industries, and has been for a long time…



Bioprocessing is now ubiquitous

TABLE 1.1 Examples of Products from Bioprocessing

Product	Typical organism used
BIOMASS	
Agricultural inoculants for nitrogen fixation	*Rhizobium leguminosarum*
Bakers' yeast	*Saccharomyces cerevisiae*
Cheese starter cultures	*Lactococcus* spp.
Inoculants for silage production	*Lactobacillus plantarum*
Single-cell protein	*Candida utilis* or *Pseudomonas methylotrophus*
Yoghurt starter cultures	*Streptococcus thermophilus* and *Lactobacillus bulgaricus*
BULK ORGANICS	
Acetone/butanol	*Clostridium acetobutylicum*
Ethanol (nonbeverage)	*Saccharomyces cerevisiae*
Glycerol	*Saccharomyces cerevisiae*
ORGANIC ACIDS	
Citric acid	*Aspergillus niger*
Gluconic acid	*Aspergillus niger*
Itaconic acid	*Aspergillus itaconicus*
Lactic acid	*Lactobacillus delbrueckii*
AMINO ACIDS	
L-Arginine	*Brevibacterium flavum*
L-Glutamic acid	*Corynebacterium glutamicum*
L-Lysine	*Brevibacterium flavum*
L-Phenylalanine	*Corynebacterium glutamicum*
Others	*Corynebacterium* spp.
NUCLEIC ACID-RELATED COMPOUNDS	
5′-guanosine monophosphate (5′-GMP)	*Bacillus subtilis*
5′-inosine monophosphate (5′-IMP)	*Brevibacterium ammoniagenes*
ENZYMES	
α-Amylase	*Bacillus amyloliquefaciens*
Glucoamylase	*Aspergillus niger*
Glucose isomerase	*Bacillus coagulans*
Pectinases	*Aspergillus niger*
Proteases	*Bacillus* spp.
Rennin	*Mucor miehei* or recombinant yeast
VITAMINS	
Cyanocobalamin (B_{12})	*Propionibacterium shermanii* or *Pseudomonas denitrificans*
Riboflavin (B_2)	*Eremothecium ashbyii*
EXTRACELLULAR POLYSACCHARIDES	
Dextran	*Leuconostoc mesenteroides*
Xanthan gum	*Xanthomonas campestris*
Other	*Polianthes tuberosa* (plant cell culture)
POLY-β-HYDROXYALKANOATE POLYESTERS	
Poly-β-hydroxybutyrate	*Alcaligenes eutrophus*
ANTIBIOTICS	
Cephalosporins	*Cephalosporium acremonium*
Penicillins	*Penicillium chrysogenum*
Aminoglycoside antibiotics (e.g., streptomycin)	*Streptomyces griseus*
Ansamycins (e.g., rifamycin)	*Nocardia mediterranei*
Aromatic antibiotics (e.g., griseofulvin)	*Penicillium griseofulvum*
Macrolide antibiotics (e.g., erythromycin)	*Streptomyces erythreus*
Nucleoside antibiotics (e.g., puromycin)	*Streptomyces alboniger*
Polyene macrolide antibiotics (e.g., candidin)	*Streptomyces viridoflavus*
Polypeptide antibiotics (e.g., gramicidin)	*Bacillus brevis*
Tetracyclines (e.g., 7-chlortetracycline)	*Streptomyces aureofaciens*
ALKALOIDS	
Ergot alkaloids	*Claviceps paspali*
Taxol	*Taxus brevifolia* (plant cell culture)
SAPONINS	
Ginseng saponins	*Panax ginseng* (plant cell culture)
PIGMENTS	
β-Carotene	*Blakeslea trispora*
PLANT GROWTH REGULATORS	
Gibberellins	*Gibberella fujikuroi*
INSECTICIDES	
Bacterial spores	*Bacillus thuringiensis*
Fungal spores	*Hirsutella thompsonii*

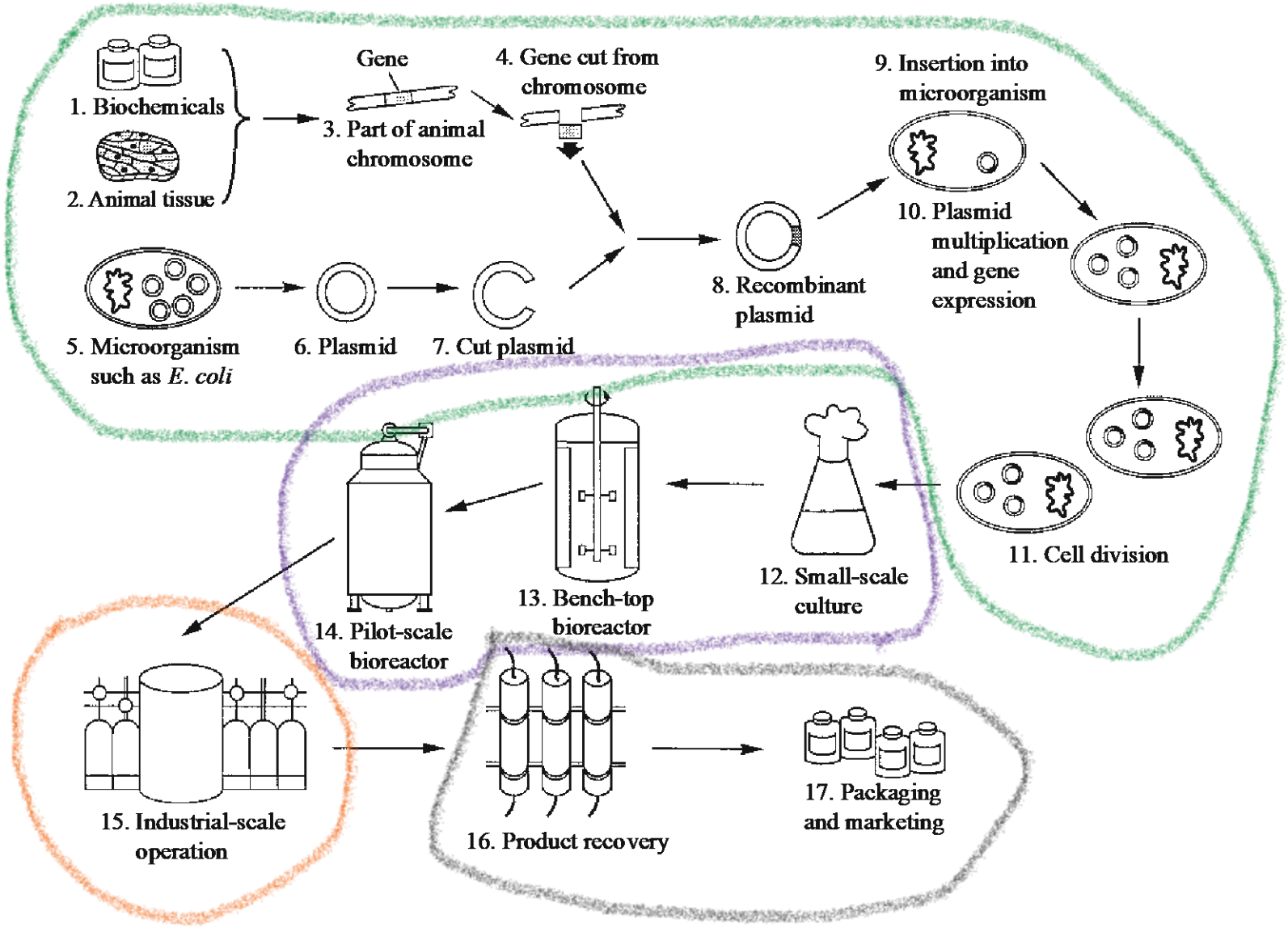
(*Continued*)

TABLE 1.1 Examples of Products from Bioprocessing (Continued)

Product	Typical organism used
MICROBIAL TRANSFORMATIONS	
D-Sorbitol to L-sorbose (in vitamin C production)	*Acetobacter suboxydans*
Steroids	*Rhizopus arrhizus*
VACCINES	
Diphtheria	*Corynebacterium diphtheriae*
Hepatitis B	Surface antigen expressed in recombinant *Saccharomyces cerevisiae*
Mumps	Attenuated viruses grown in chick embryo cell cultures
Pertussis (whooping cough)	*Bordetella pertussis*
Poliomyelitis virus	Attenuated viruses grown in monkey kidney or human diploid cells
Rubella	Attenuated viruses grown in baby hamster kidney cells
Tetanus	*Clostridium tetani*
THERAPEUTIC PROTEINS	
Erythropoietin	Recombinant mammalian cells
Factor VIII	Recombinant mammalian cells
Follicle-stimulating hormone	Recombinant mammalian cells
Granulocyte–macrophage colony-stimulating factor	Recombinant *Escherichia coli*
Growth hormones	Recombinant *Escherichia coli*
Hirudin	Recombinant *Saccharomyces cerevisiae*
Insulin and insulin analogues	Recombinant *Escherichia coli*
Interferons	Recombinant *Escherichia coli*
Interleukins	Recombinant *Escherichia coli*
Platelet-derived growth factor	Recombinant *Saccharomyces cerevisiae*
Tissue plasminogen activator	Recombinant *Escherichia coli* or recombinant mammalian cells
MONOCLONAL ANTIBODIES	
Various, including Fab and Fab_2 fragments	Hybridoma cells
THERAPEUTIC TISSUES AND CELLS	
Cartilage cells	Human (patient) chondrocytes
Skin	Human skin cells

Non-exhaustive list from Chapter 1 of ``Bioprocess Engineering Principles''. No need to remember this

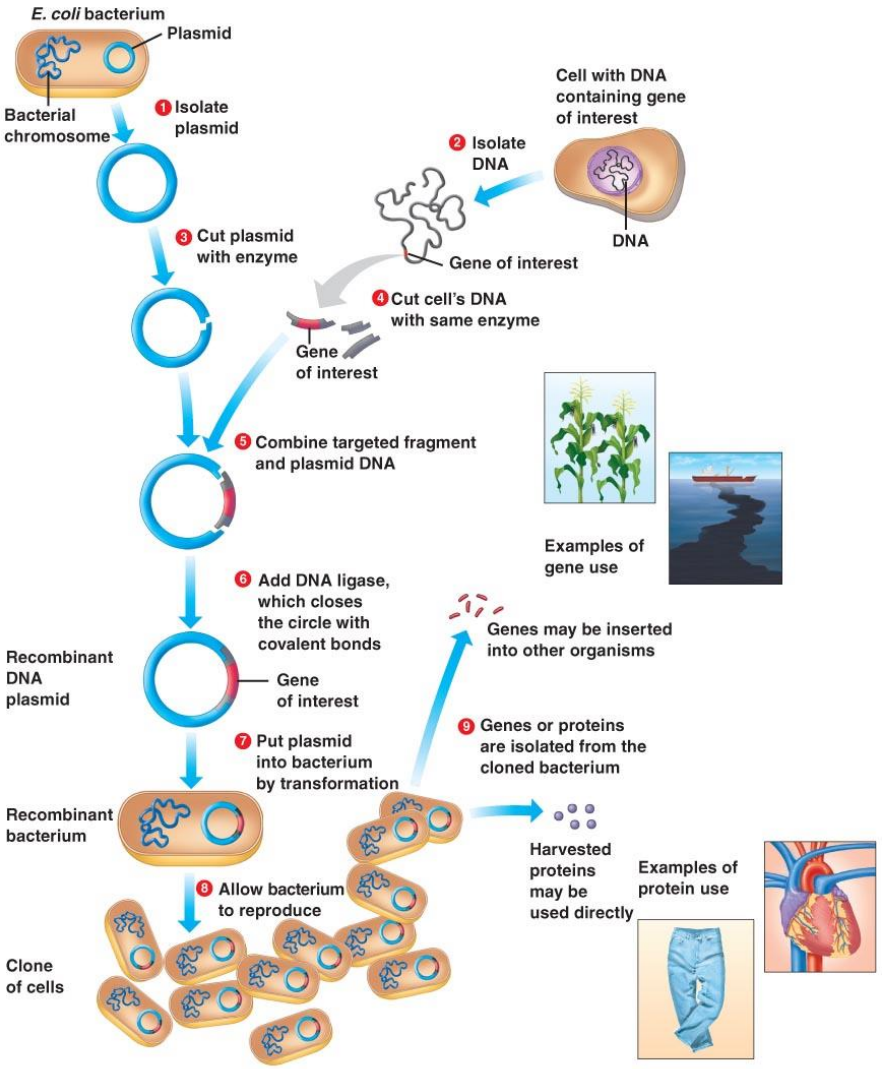
Developing a bioprocess



Chapter 1 of ``Bioprocess Engineering Principles''

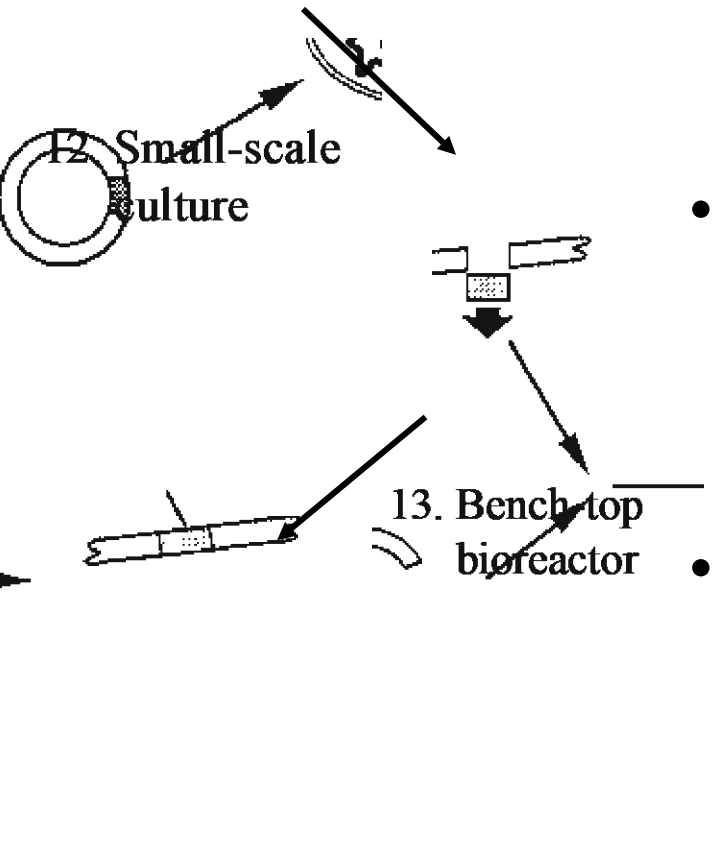
- Steps 1-11: developing the micro-organism
- Steps 12-14: scaling up the culture
- Step 15: large-scale fermentation (upstream processing)
- **Steps 16-17: Product extraction and purification (downstream processing)**

Developing a microorganism for bioprocessing



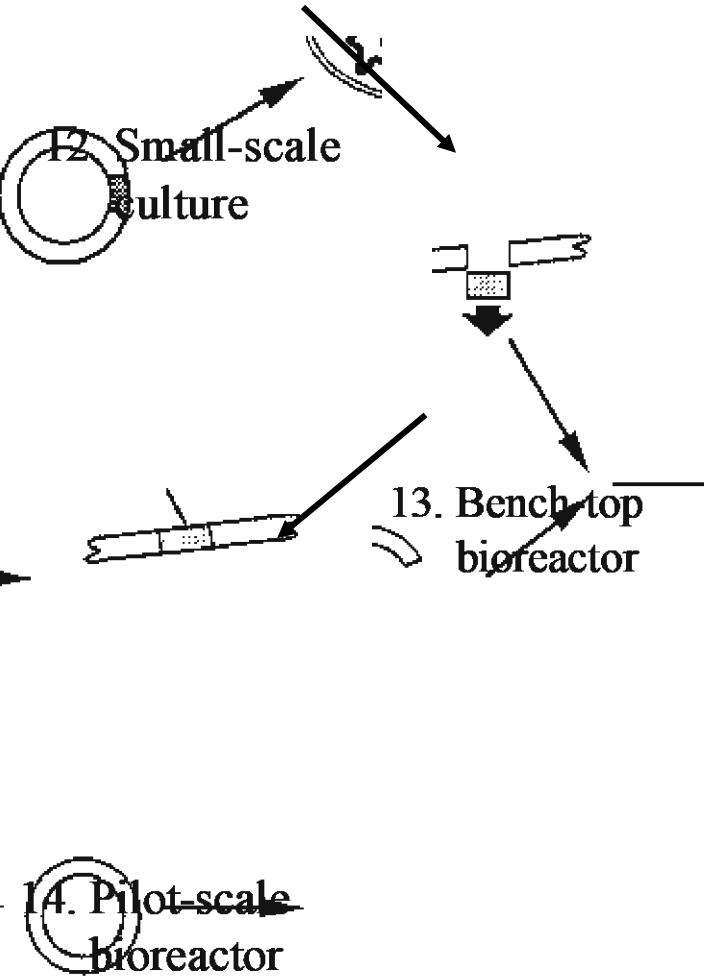
- Example is for a recombinant product (protein) that can be encoded in DNA
- Gene of interest is identified in a cell, and extracted
- Gene is cloned into a plasmid
- Cells (in the example *E. coli*) are transformed with the plasmid, so they produce the target recombinant protein
- Cell used is chosen for its ability to produce efficiently the target product, but also based on how easy it is to culture at scale, harvest etc. Engineering considerations made at this stage
- **You have learned all about this!**

Scaling up



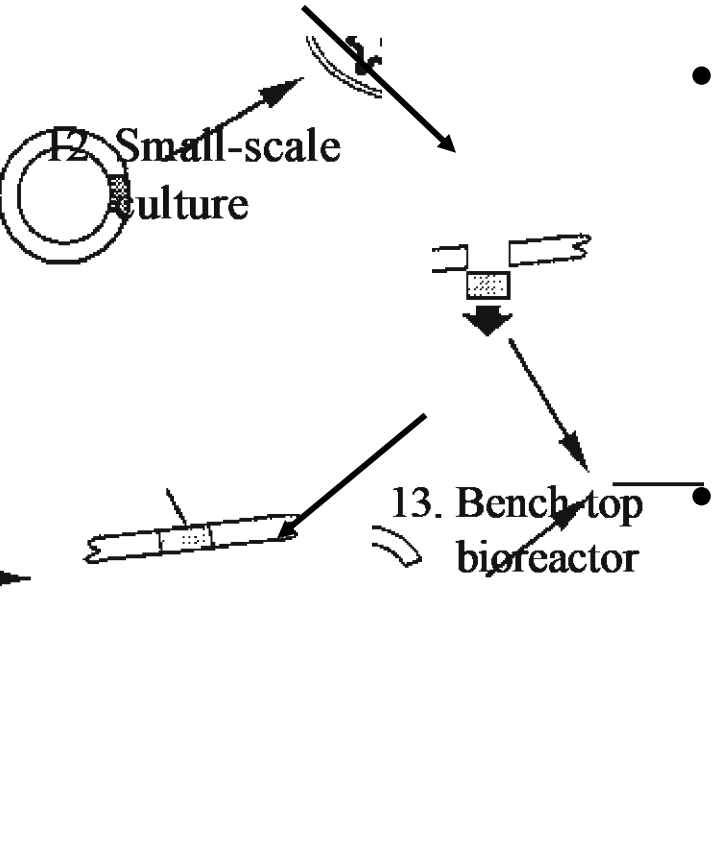
- Microorganism development is done primarily in very small cultures of a few hundreds of microliters (well-plates), followed by **small-scale cultures** in flasks (250 ml to 1l)
- At this stage, medium composition, pH, temperature, and other environmental conditions allowing optimal growth and productivity are determined
- The performance of the organism is determined using parameters such as growth rate, and product yield

Scaling up



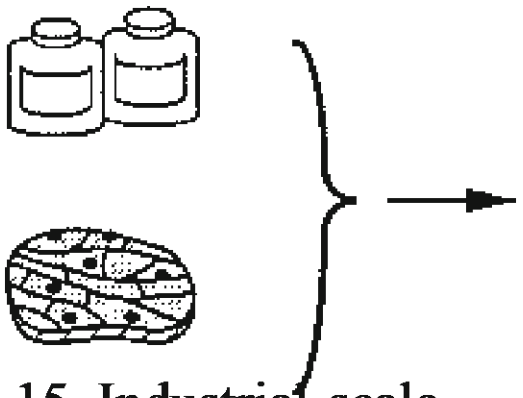
- Once growth parameters are known, scaling up starts with a **bench-top bioreactor** (1-2l)
- In the bioreactor we can control and measure parameters such as **stirrer speed, dissolved oxygen concentration, pH and temperature**.
- Better control compared to flasks, allows better characterization of optimal growth conditions and collecting information on **shear sensitivity, oxygen requirements, foaming etc**.
- Important process parameters can be measured or computed, including **oxygen transfer rate, power number, Reynolds number, mixing time etc**.

Scaling up



- We then move to a **pilot-scale bioreactor** (100-1000l)
- Usually this is geometrically similar to the bench-top reactor. Rates of air sparging and stirring may change to enable sufficient oxygen transfer and fit power requirements
- Changes, often loss, of productivity may occur and need to be assessed at this stage

Upstream processing



15. Industrial-scale operation

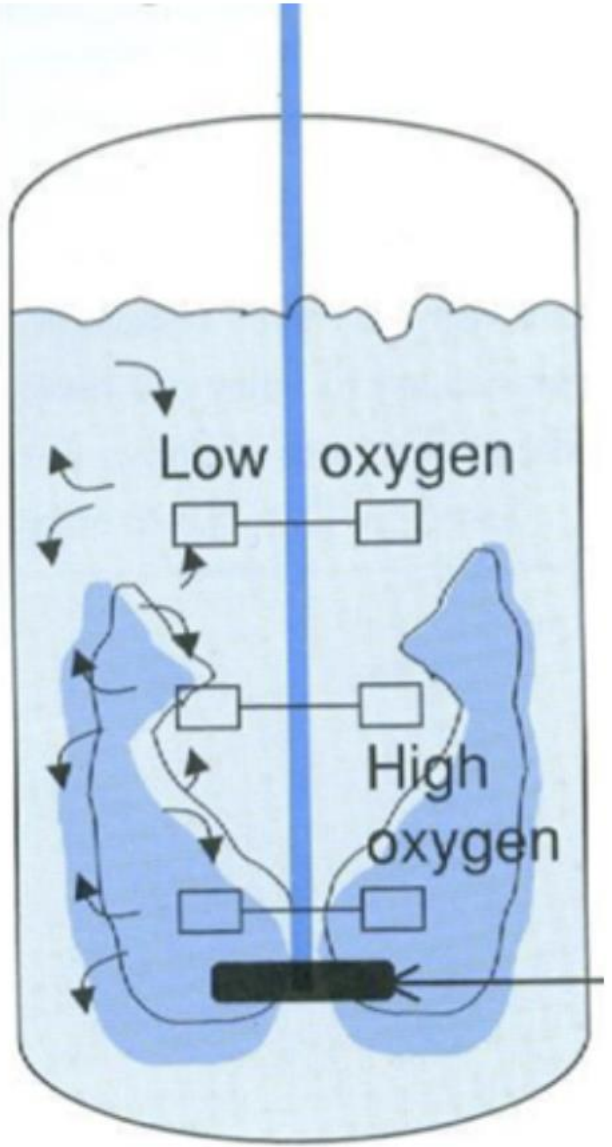
- If pilot-scale testing is successful, one proceeds to designing and **industrial scale plant**, where many considerations around auxiliary facilities need to be made (air supply, sterilization, cooling water supply, medium preparation),
- Preparation and sterilization of media
- Fermentation or cell-free bioprocesses conducted in the designed bioreactors
- Bioreactors designed considering requirements for (gentle) mixing, oxygenation, temperature, foam reduction, energy efficiency.
- **You have learned about this in the reactors courses**

Lecture outline

1) Overview of bioprocessing
2) **(More detailed) overview of downstream processing**
3) Principles of filtration for cell removal
4) Mathematical description of dead-end filtration

Downstream processing

- 10 – 300 m^3 of fermentation broth
- Mainly water – the fraction of product could be high, but also as low as *parts per billion*
- Incredibly complex mixtures of cells, proteins, sugars and other macromolecules, many of which are chemically similar to the desired product
- **We need to extract and purify the product!**



Downstream processing

The treatment of culture broth after completion of fermentation is known as downstream processing

- Downstream processing is typically concerned with **separation and concentration** of the desired product
- Typically, no further chemical transformations occur
- Separations in a biological context can be challenging for a variety of reasons (next slide)
- Can account for **the major proportion of the total production cost**

Challenges with downstream processing

- Fermentation **products are often formed in dilute solution** – both cell culture media and cells comprise mainly of water
- Biological **products are labile** – this restricts the suitable range of temperatures, pressures, solvents and salt concentrations for product recovery
- Bio **products are susceptible to contamination** – it is usually not feasible to maintain aseptic conditions throughout the entire process plant
- Downstream processing must be **quick and efficient to avoid product degradation**

Downstream processing separation stages

Separation techniques in bioprocessing can be classified into the following **five stages:**

1. **Cell removal**: this is commonly the first step after fermentation. Products could be within the cell, outside the cell, or the cells may be the product. In any case, separating the cells from the fermentation media is a major step in product recovery. Typical unit operations comprise **filtration** or **centrifugation.** Covered in this course.

2. **Cell disruption**: when the products are intra-cellular, cells must be lysed. This can be achieved using **mechanical**, **physical (non-mechanical)**, **chemical** or **enzymatic methods**. The remaining cell debris must then be removed using the aforementioned cell removal methods. Covered in this course**.**

Downstream processing separation stages

3. **Primary isolation:** the aim of the primary isolation stage is to remove components with substantially different properties to the product of interest. Primary isolation processes are usually non-selective and depend on the chemical and physical properties of the product. Typical operations include **solvent extraction**, **two-phase liquid extraction**, **adsorption**, **precipitation** (covered in this course) or **ultrafiltration** (done in separations)

4. **Product enrichment:** these are highly selective methods that separate products from impurities with similar physical or chemical properties. Different types of **chromatography** are commonly used for product enrichment (covered in this course).

5. **Final isolation:** the final form of the product will vary depending on the application. Generally, liquid products are processed via **ultrafiltration** and solid products **crystallised** before **centrifugation** or **filtration** and **drying** (done in this separations).

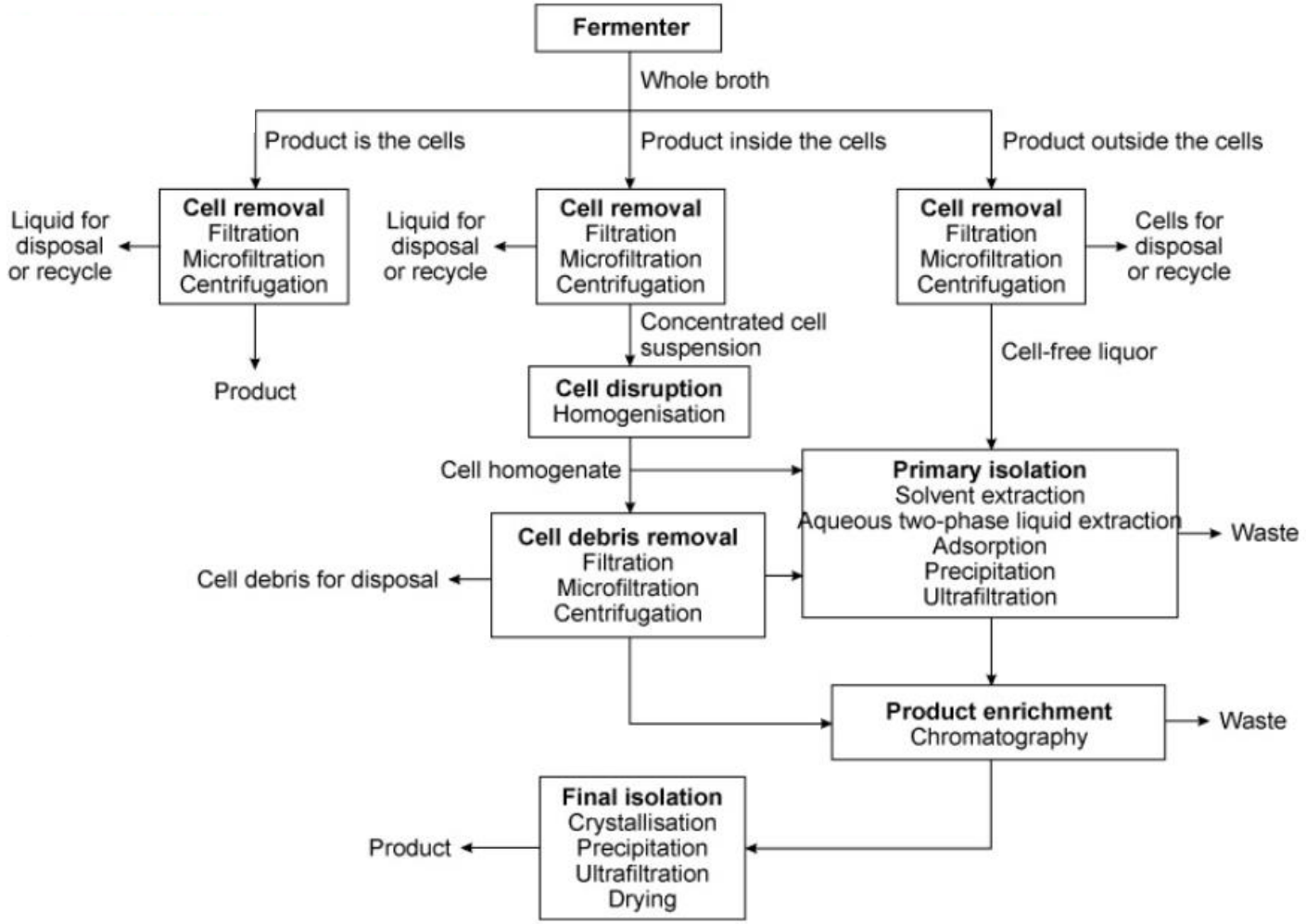
Downstream processing separation stages

At each step, product concentration and purity increase steeply

Step	Product concentration (g l^{-1})	Product quality (%)
Harvest broth	0.1 – 5	0.1 – 1
Cell removal	1 – 5	0.2 – 2
Primary isolation	5 – 50	1 – 10
Product enrichment	50 – 200	50 – 80
Final isolation	50 – 200	90 – 100

Downstream process logic

Process to be tailored to specific needs, but process logic can be generalised as shown in below

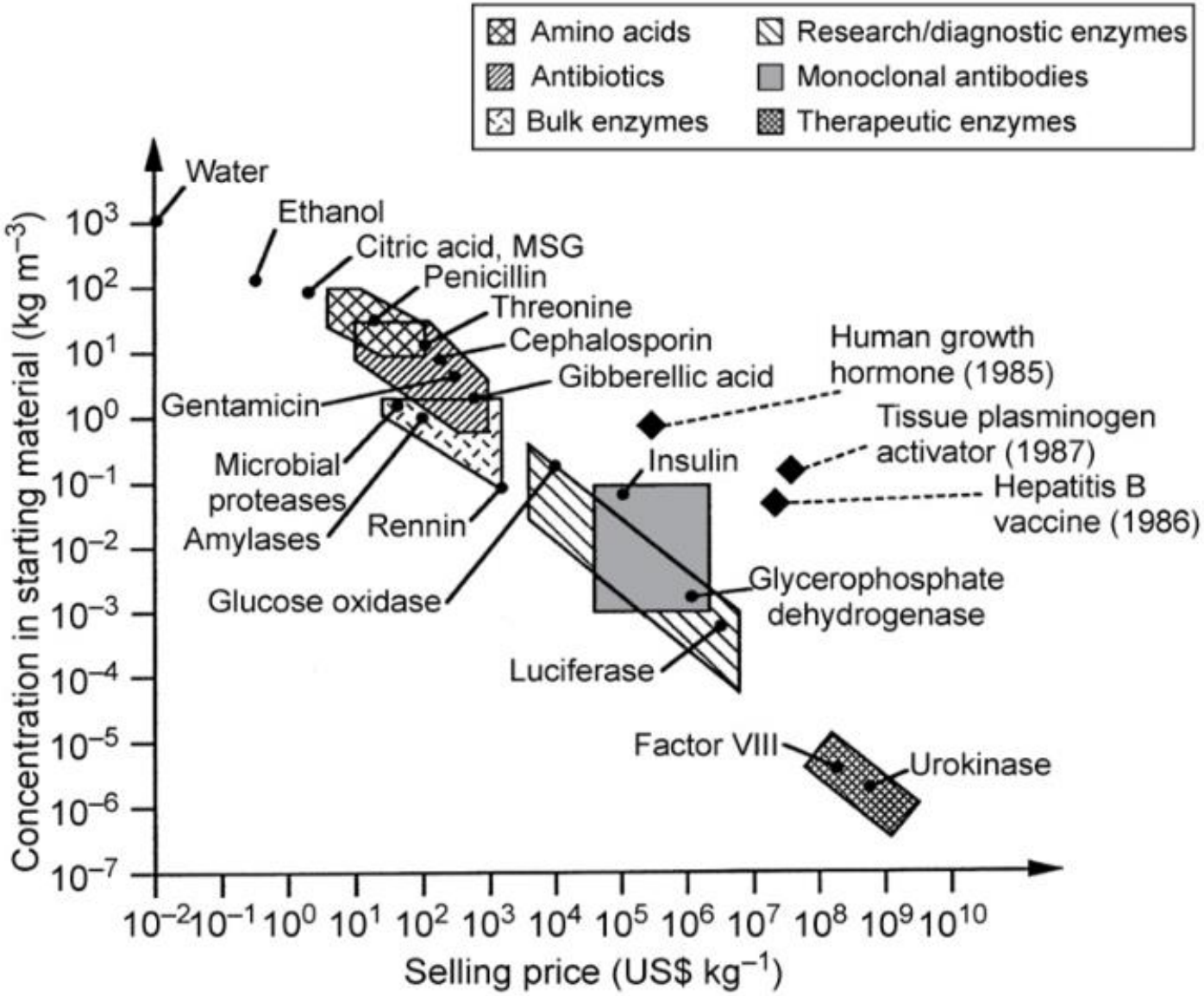


Cost of downstream processing

- Downstream processing can account for a substantial proportion of the total production cost of a fermentation product

- The ratio of fermentation cost to product recovery cost for well characterised antibiotics such as penicillin is 60:40, however, for many bio products, such as recombinant proteins or monoclonal antibodies, this ratio is reversed and can be up to 20:80

Cost of downstream processing



- For certain compounds initial concentration is very low, which increases recovery costs and thus selling price
- This leaves an open ground for innovation
- **Chemical engineers have the potential to level the cost scale in many modern biological processes**

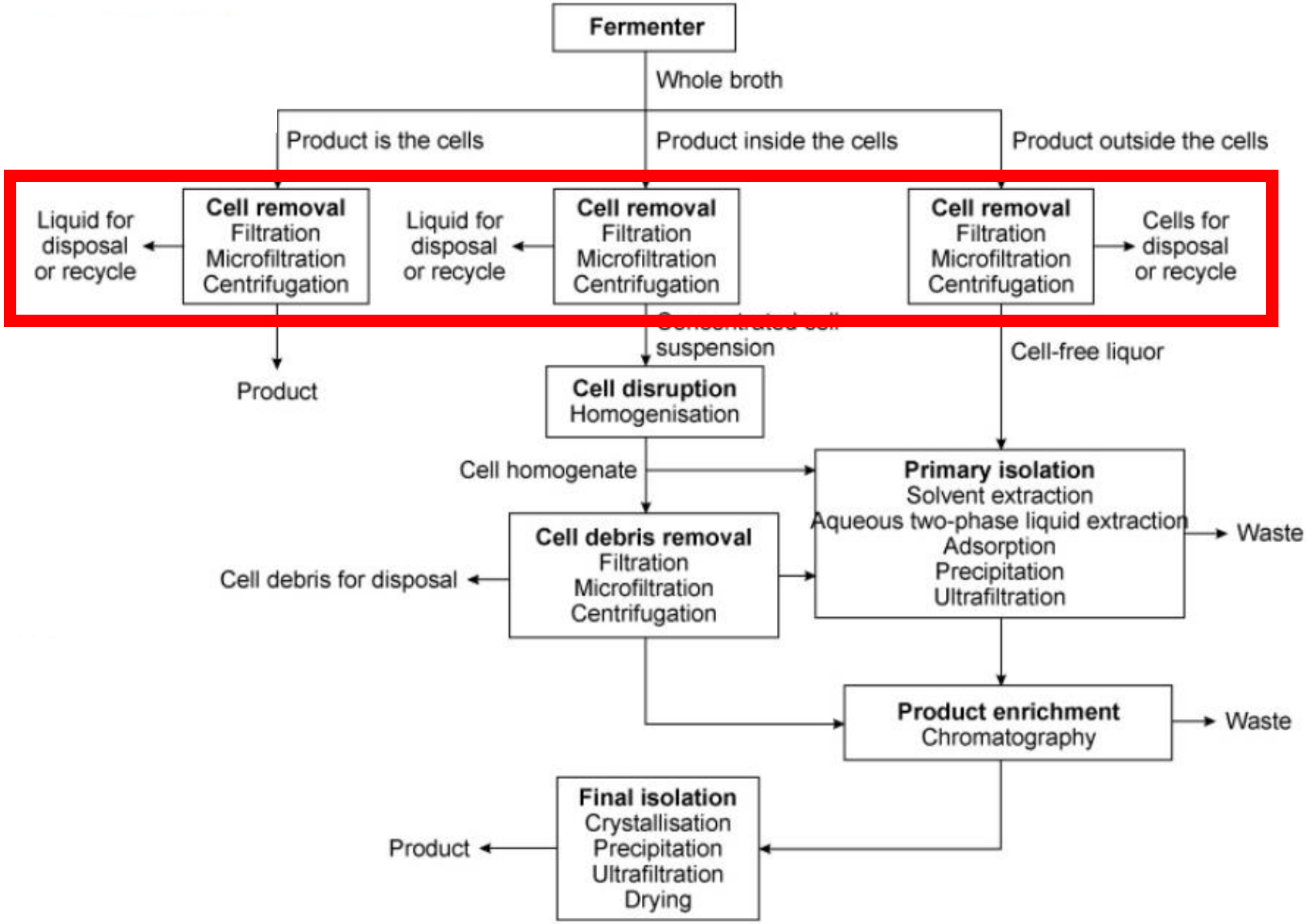
Lecture outline

1) Overview of bioprocessing
2) (More detailed) overview of downstream processing
3) **Principles of filtration for cell removal**
4) Mathematical description of dead-end filtration

Cell removal

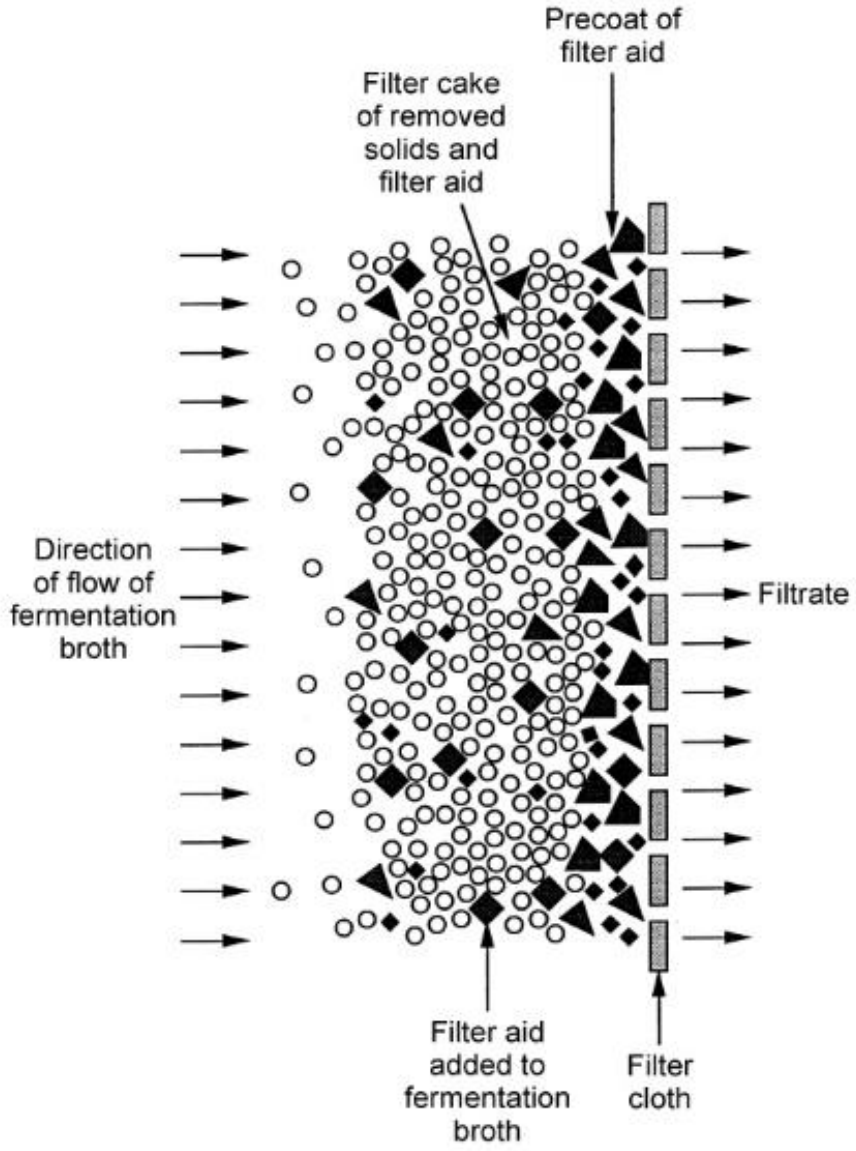
- Cell removal unit operations essentially separate particulates from the fermentation broth or other liquids
- **Filtration** and **centrifugation** procedures achieve this by exploiting particle size and density, respectively
- In some cases, the fermentation broth is pre-treated to improve the efficiency of separation, *e.g.* by including additives that reduce viscosity and/or promote cell aggregation
- In this lecture we introduce **filtration**, covering the basics of design and operation as applied to bioprocessing.
- **Centrifugation** will be examined in Lecture 10

Downstream process logic

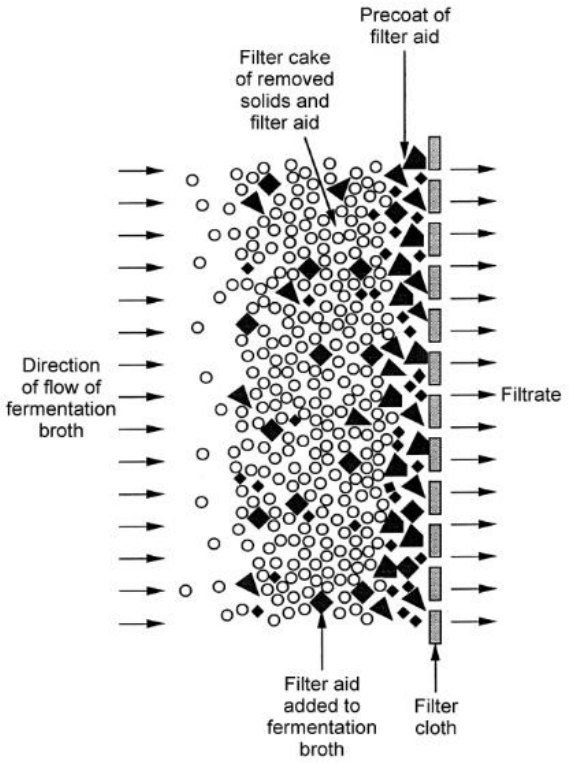


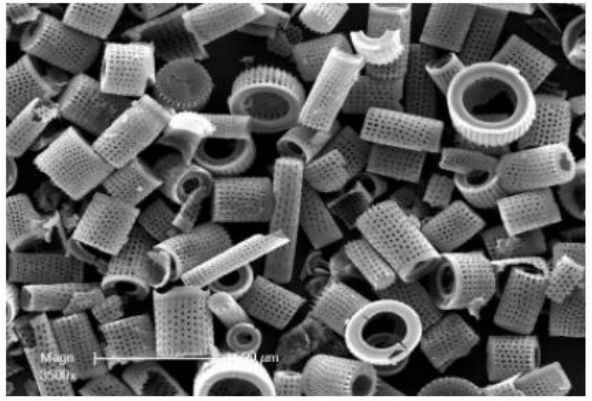
Dead-end filtration

- 'Dead-end' filtration is perhaps the simplest form of solid-liquid separation
- Simple principle; a physical barrier with openings smaller than the suspended particles impedes motion of the latter, whereas the liquid can flow past the particles resulting in the separation of the two phases
- This process can be promoted by gravity or by pressure drop



Pre-coating filters and filter aids



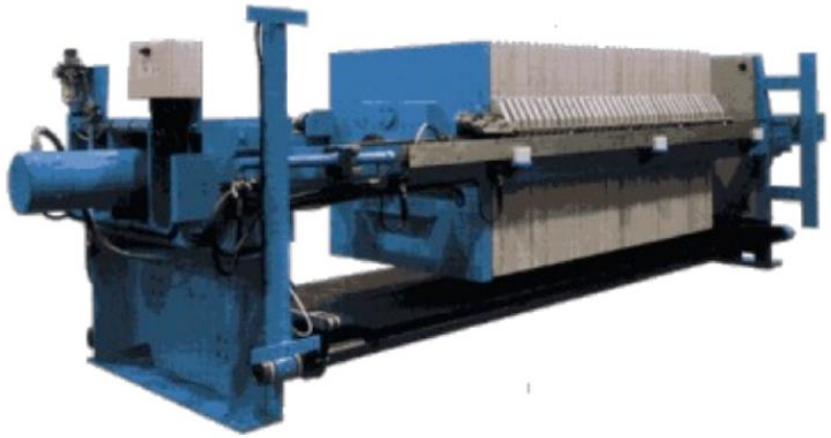


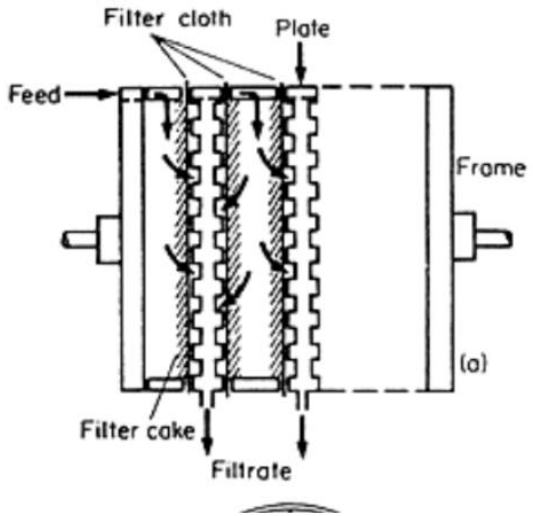
- Filter is usually a "filter cloth"
- A **filter cake** is progressively formed on the cloth. The cake aids filter efficiency, but also increases flow resistance
- **Filter aids**, e.g. Kieselguhr particles can be added to the broth. These are naturally occurring materials used to make the cake more porous and decrease resistance
- The cloth can be pre-coated with filter aid particles

Issues with dead-end filtration

- The major problem with dead-end filtration is cake formation during the process
- As the cake forms, it deteriorates the performance of the filter by imposing a greater resistance to liquid flow
- Eventually, the liquid flow will drop to zero if the pressure difference is not increased
- To overcome this issue, filter aids are used, e.g. Kieselguhr particles can be added to the broth or used to precoat the filter in order to increase cake porosity, as mentioned earlier

Batch filtration: plate and frame filter





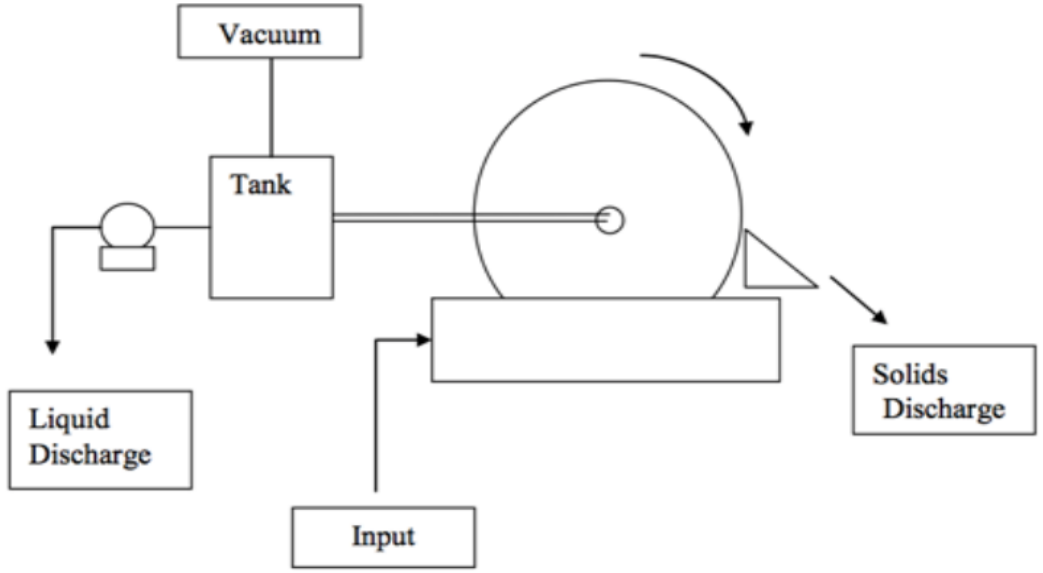
- Cloth spread out over plates, which support the cloth along ridges leaving at the same time a free area below the cloth for flow of the filtrate.
- Larger number of plates gives sufficient area
- Cake builds up on the upstream side of the cloth
- When available space between plates if filled with cake, the filter press needs to be dismantled and scraped off

Issues with batch dead-end filtration

- Conventional plate filters are also laborious to operate, as the filter must be frequently serviced for the removal of the filter cake
- This makes them suitable for small batch fermentation processes
- For large fermentation processes, **continuous rotary drum vacuum filters** are more commonly used

Continuous rotary drum vacuum filter

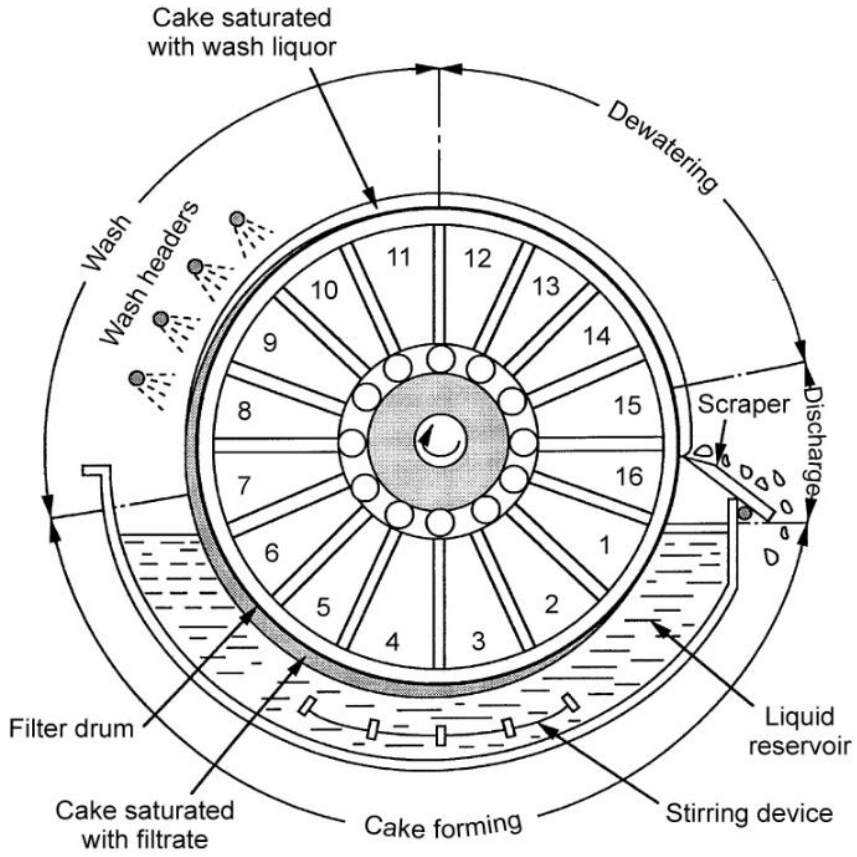




- Commonly applied for dead-end filtration
- Can process large volumes
- Ideal for separation of organisms
- Difficult to operate aseptically

Continuous rotary drum vacuum filter

- Horizontal cylindrical drum of 0.5 to 3 m in diameter is covered with filter cloth and rotated slowly at 0.1 to 2 rpm
- A vacuum is applied inside the drum, drawing liquid from the immersed portion to the interior of the drum
- As the drum rotates out of the reservoir, the filter cake is sprayed with wash liquid
- Continued application of vacuum results in the cake being dewatered before removal with a scraper
- The liquid is drawn through the cloth and collected in a separate tank



Cheap and easy but not aseptic

Lecture outline

1) Overview of bioprocessing
2) (More detailed) overview of downstream processing
3) Principles of filtration for cell removal
4) **Mathematical description of dead-end filtration**

Dead-end filtration theory

Dead-end filtration is governed by the following differential equation:

$$\frac{1}{A}\frac{dV_f}{dt} = \frac{\Delta p}{\mu_f \left[\phi\left(\frac{M_c}{A}\right) + R_M\right]}$$

A is the filter area (key parameter)

V_f is the filtrate volume

t is the filtration time

Δp is the pressure drop across the filter

μ_f is the filtrate viscosity

M_c is the total mass of solids in the cake

ϕ is the average specific cake resistance

R_M is the filter medium resistance.

Derivation (non examinable): Ripperger et al. 2013 https://doi.org/10.1002/14356007.b02_10.pub3

Dead-end filtration theory

Generally, we can assume that $M_c = cV_f$, where ***c* is the mass of solids deposited per unit volume of the filtrate**. The equation then simplifies to

$$\frac{1}{A}\frac{dV_f}{dt} = \frac{\Delta p}{\mu_f \left[\phi\left(\frac{cV_f}{A}\right) + R_M\right]}$$

The equation can be rearranged as

$$dt = \frac{\mu_f}{A\Delta p}\left(\frac{\phi c}{A}V_f + R_M\right)dV_f$$

We can integrate between time zero and t and volume zero and V_f

$$t = \frac{\mu_f}{A\Delta p}\left(\frac{\phi c}{2A}V_f + R_M\right)V_f$$

Dead-end filtration theory

This equation gives the **filtration time** needed to filter a volume V_f.

$$t = \frac{\mu_f}{A\Delta p}\left(\frac{\phi c}{2A}V_f + R_M\right)V_f$$

Generally, the filter medium resistance is R_M is negligible compared to the resistance coming from the cake $\frac{\phi c}{2A}V_f$, so we can assume $R_M \ll \frac{\phi c}{2A}V_f$ and simplify as

$$t = \frac{\phi\ c\ \mu_f\ V_f^2}{2\ A^2\ \Delta p}$$

Dead-end filtration theory

$$t = \frac{\phi\, c\, \mu_f\, V_f^2}{2\, A^2\, \Delta p}$$

We can further note that the specific cake resistance ϕ is related to the pressure drop Δp as (derivation not required)

$$\phi = \phi' \Delta p^s$$

Where $\boldsymbol{\phi'}$ **is a constant** dependent on the size and morphology of the particles in the cake, while $\boldsymbol{s}$ **is the cake compressibility**. Substituting in the equation above we obtain

$$t = \frac{\phi' c\, \mu_f\, V_f^2}{2\, A^2 \Delta p^{1-s}}$$

Dead-end filtration theory

$$t = \frac{\phi' c\, \mu_f\, V_f^2}{2\, A^2 \Delta p^{1-s}}$$

A is the filter area (key parameter)

V_f is the filtrate volume

t is the filtration time

Δp is the pressure drop across the filter

μ_f is the filtrate viscosity

s is the cake compressibility

ϕ' is a constant that depends on particle size, shape etc.

c is the mass concentration of solids

Valid under the assumption that the filter medium resistance, R_M, is negligible compared to the cake resistance

Exercise: Filtration of mycelial broth

A $30\ \mathrm{ml}$ sample of broth from a penicillin fermentation is filtered in the laboratory on a $3\ \mathrm{cm}^2$ filter at a pressure drop of $0.35\ \mathrm{bar}$

The filtration time is $4.5\ \mathrm{min}$ Previous studies have shown that *Penicillium chrysogenum* filter cake is compressible with $s = 0.5$

If $500\ l$ of broth from a pilot-scale fermenter must be filtered in 1 hour, what area of filter is required if the pressure drop is:

a) $0.7\ \mathrm{bar}$, and b) $0.35\ \mathrm{bar}$

The resistance due to the filter medium is negligible

Solution

Given that the R_M is negligible, we can use

$$t = \frac{\phi' c \, \mu_f \, V_f^2}{2 \, A^2 \Delta p^{1-s}}$$

Rearranging:

$$A = \sqrt{\frac{\phi' c \, \mu_f \, V_f^2}{2t\Delta p^{1-s}}}$$

To compute the area, we first need to work out $\phi' c \, \mu_f$ from the information given on the lab sample

Solution

$$\phi' c\, \mu_f = \frac{t\ 2A^2 \Delta p^{1-s}}{V_f^2}$$

Substituting the values for the small-scale filtration:

$$\phi' c\, \mu_f = \frac{4.5 \times 60\ \mathrm{s}\ \times 2 \times (3 \times 10^{-4} \mathrm{m}^2)^2 (0.35 \times 10^5 \mathrm{Pa})^{0.5}}{(3 \times 10^{-5} \mathrm{m}^3)^2}$$

$$= 10^7 \mathrm{s\ m^{-2} Pa^{0.5}}$$

Solution

We can now compute the areas For point a) $\Delta p = 0.7\ \text{bar} = 0.7 \times 10^5 \text{Pa}$

$$A = \sqrt{\frac{\phi' c\ \mu_f\ V_f^2}{2t}}\,\Delta p^{(s-1)/2}$$

$$A = \sqrt{\frac{10^7 \text{s m}^{-2} \text{Pa}^{0.5} (0.5\ \text{m}^3)^2}{2 \times 3600\ \text{s}}}\,(0.7 \times 10^5)^{-0.25} = 1.15 \text{m}^2$$

Solution

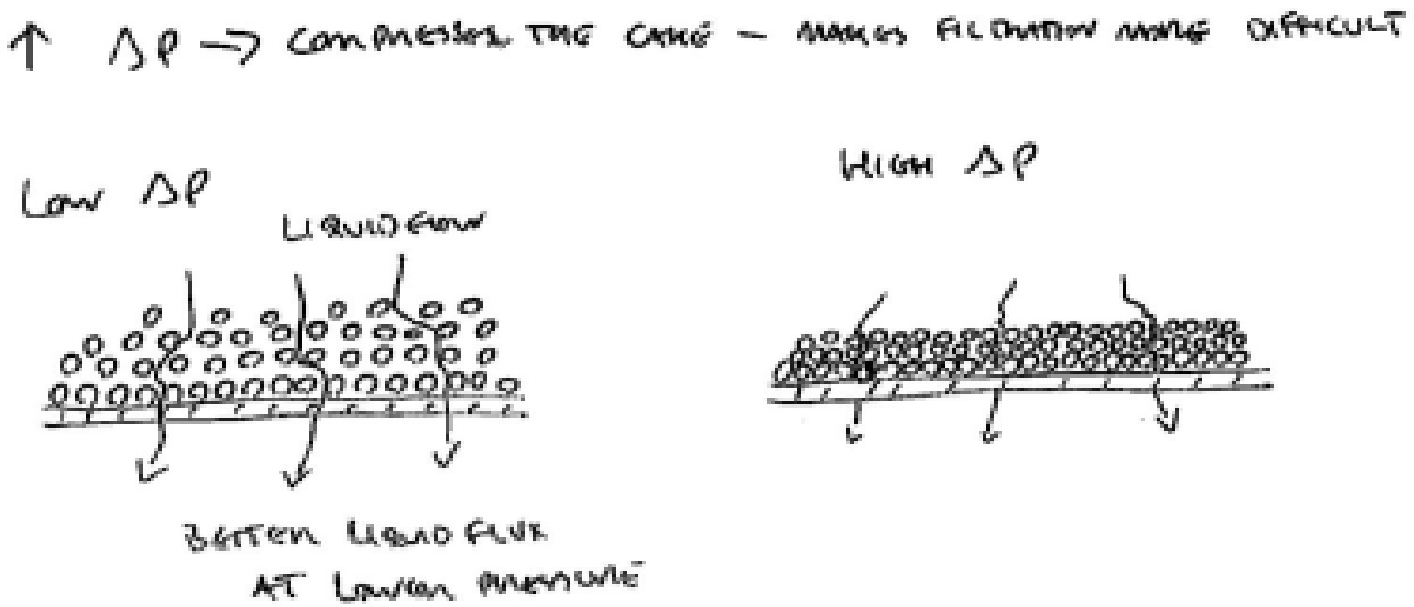
For point b) $\Delta p = 0.35\ \text{bar} = 0.35 \times 10^5 \text{Pa}$

$$A = \sqrt{\frac{\phi' c\ \mu_f\ V_f^2}{2t}}\,\Delta p^{(s-1)/2}$$

$$A = \sqrt{\frac{10^7 \text{s m}^{-2} \text{Pa}^{0.5} (0.5\ \text{m}^3)^2}{2 \times 3600\ \text{s}}}\,(0.35 \times 10^5)^{-0.25} = 1.36 \text{m}^2$$

Solution

Halving the pressure drop only increases the required area by 20%. Why?



1) Mathematically: weak dependence of area on Δp

2) Physically: Increasing pressure compresses cake making filtration more difficult!